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Research article

Identification and expression analysis of the NAC transcription factor family in durum wheat (*Triticum turgidum* L. ssp. *durum*)

Mohammed Najib Saidi*, Dhawya Mergby, Faiçal Brini

Biotechnology and Plant Improvement Laboratory, Centre of Biotechnology of Sfax, PO Box 1177, Road Sidi Mansour 6 km, Sfax 3018, Tunisia

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The NAC (NAM, ATAF and CUC) proteins belong to one of the largest plant-specific transcription factor (TF) families and play important roles in plant development processes, response to biotic and abiotic cues and hormone signaling. Our analysis led to the identification of 168 NAC genes in durum wheat, including nine putative membrane-bound TFs and 48 homeologous genes pairs. Phylogenetic analyses of TtNACs along with their Arabidopsis, grape, barley and rice counterparts divided these proteins into 8 phylogenetic groups and allowed the identification of TtNAC-A7, TtNAC-B35, TtNAC-A68, TtNAC-B69 and TtNAC-A43 as homologs of OsNAC1, OsNAC8, OsNTL2, OsNTL5 and ANAC025/NTL14, respectively. *In silico* expression analysis, using RNA-seq data, revealed tissue-specific and stress responsive TtNAC genes. The expression of ten selected genes was analyzed under salt and drought stresses in two contrasting tolerance cultivars. This analysis is the first report of NAC gene family in durum wheat and will be useful for the identification and selection of candidate genes associated with stress tolerance.

1. Introduction

Plants are constantly being exposed to abiotic stresses such as drought, high salinity, threshold temperatures and nutrient deficiency. These environmental stress factors negatively affect growth and productivity. Plants possess various mechanisms to cope with stress conditions. One of them is transcription factors (TFs), these protein regulate gene expression through binding to either promoter or enhancer region of a gene. Moreover, transcription factors regulate the gene expression by a modular structure with one or more DNA-binding domains (Tombuloglu et al., 2013). Among them, NAC transcription factors are one of the largest families of transcriptional regulators in plants. Members of the NAC gene family have been suggested to play important roles in the regulation of the transcriptional reprogramming associated with plants stress responses. Members of this family are identified by the presence of the NAC domain which consists of five sub-domains (A to E) that make up motifs for both DNA-binding and protein-protein interactions (Ernst et al., 2004). A typical NAC transcription factor has the conserved NAC domain in the N-terminal (Ooka et al., 2003) as well as a more variable, transcriptional activation or repression region in the C-terminal (Hao et al., 2011). NAC proteins may have an atypical structure containing only the NAC domain (Shen et al., 2009) or have two NAC domains in tandem (Jensen et al., 2010).

Compared with Arabidopsis, rice, and other monocots species, there have been fewer investigations of NAC transcription factors in durum wheat. In rice, 151 non-redundant NAC genes were identified (Nuruzzaman et al., 2010). Several whole-genome expression profiling studies in Arabidopsis and rice reported that NAC genes were induced by at least one type of abiotic stress like salinity, drought, cold or hormone (Fang et al., 2008). Expression analysis using RNA-seq in rice revealed the induction of at least 45 and 26 NAC genes under abiotic and biotic stresses, respectively (Nuruzzaman et al., 2010). Arabidopsis NAC genes ANAC019, ANAC055 and ANAC072 were proven to enhance plant tolerance to drought stress by binding to the ERD1 promoter (Tran et al., 2004). AtNAC2 is involved in response to ethylene and auxin signaling and also involved in the response to salt stress (He et al., 2005). In rice, the over-expression of OsNAC1, OsNAC2, OsNAC5, OsNAC045, and OsNAC063 enhance the tolerance to multiple abiotic stresses in transgenic plants (Zheng et al., 2009; Takasaki et al., 2010). The bread wheat genome contains multiple NAC genes with diverse expression patterns and roles in the responses to environmental factors. Moreover, six genes encoding NAC TFs in bread wheat were

* Corresponding author.
E-mail addresses: saiidimn@gmail.com, mohamednajib.saidi@cbs.rnrt.tn (M.N. Saidi).

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identified (Tang et al., 2012). Transgenic tobacco plants over-expressing TaNAC2a showed higher fresh and dry weights compared to wild type plants under drought condition (Tang et al., 2012). Moreover, two NAC genes (GRAB1 and GRAB2) are involved in the resistance to wheat dwarf virus (Xie et al., 1999). TaNAM-B1 controls the distribution of nutrients between leaves and developing grains and modulate grain protein, zinc, and iron contents (Uauy et al., 2006). The expression of TaNAC2 significantly increased in response to salt, low temperature, and drought stresses and its enhanced the tolerance to abiotic stresses (drought, saltand freezing) in transgenic Arabidopsis by the activation of several stress-responsive genes (Mao et al., 2012). Furthermore, the over-expression of wheat TaNAC2L (close homoeolog of TaNAC2) is induced in wheat under high temperature (40 °C). It over-expression in *Arabidopsis thaliana* enhanced acquired heat tolerance and also activated the expression of heat-related genes (Guo et al., 2015). Moreover, TaNAC69 regulates several stress-associated genes enhancing the adaptation to drought stress (Xue et al., 2011). TaNAC67 responds to water deficit, high salinity, low temperature, and ABA treatments, and its over-expression in Arabidopsis significantly enhanced the tolerance to drought, salt, and freezing stresses (Mao et al., 2014). As far as we know, only a few NAC genes were characterized in durum wheat (*Triticum turgidum* ssp. *durum*). Indeed, TtNAM-B1 was shown to regulate senescence and improves grain protein, zinc, and iron content in transgenic wheat (Uauy et al., 2006). Besides, TtNAM-B2 is involved in cellular responses to osmotic stress (Baloglu et al., 2012).

Genome and transcriptome-wide approaches allowed the identification of NAC TF families in several plant species (Shen et al., 2009). Completion of the high-quality sequencing of the durum wheat transcriptome (Krasileva et al., 2013) has provided an excellent opportunity for transcriptome-wide analysis of all the genes belonging to specific gene families. Here, we provide a more extensive knowledge on durum wheat NAC transcription factors by the characterization of 168 members, 95 of them representing full-length coding sequences. Furthermore, the *in silico* expression profiling using RNA-seq data revealed that the expression of several TtNAC genes was modulated by abiotic stresses. The expression of ten candidate genes was carried in leaf and root of two durum wheat cultivar with contrasting tolerance to salt and drought stresses. The results provide good evidence that, in durum wheat, the NAC gene family is involved in regulatory pathways and some members could be evaluated as candidate genes to improve the durum wheat tolerance to drought and salinity.

2. Material and methods

2.1. Identification and phylogenetic analysis of NAC superfamily in durum wheat

For the identification of NAC proteins, *T. turgidum* proteins (66.633 proteins) translated from sequenced transcriptome which were generated from multiples tissues (root, shoot and grain) of 2–3 weeks old plants (Krasileva et al., 2013) were retrieved and analyzed with iTAK software v1.6 which allows the identification of plant transcription factors from protein or nucleotide sequences (Pearce et al., 2015). The presence of specific NAM domain (PF02365) in the resulting protein sequences was determined using the Pfam database (http://pfam.sanger.ac.uk). For phylogenetic analysis of the durum wheat NAC protein sequences were aligned with ClustalW and then MEGA v5.0 software (Tamura et al., 2011) was used to generate the unrooted phylogenetic tree which was constructed by the Neighbor-Joining method with a bootstrap test (1000 replicates). Barley (HNAC) NAC proteins (Christiansen et al., 2011) and NAC sequences for rice (ONAC), Arabidopsis (AtNAC) and grape (VvNAC) were retrieved as described (Shen et al., 2009). The prediction of putative membrane-bound NAC proteins was performed using the TMHMM server v.2.0 (http://www.cbs.dtu.dk/services/TMHMM/).

2.2. Determination of conserved protein motifs

Conserved motifs were investigated by multiple alignment analyses using Multiple Expectation Maximization for Motif Elicitation (MEME) v.4.7.0 (Bailey et al., 2006) with following parameters: distribution of motifs, zero or one per sequence; maximum number of motifs, 20; minimum number of sites, 2; maximum number of sites was 168; minimum motif width, 6; maximum motif width, 50.

2.3. Chromosomal location and homeologous genes identification

To determine the approximate chromosomal location of NAC genes, cDNA sequences of TtNACs were subjected to BLASTN analysis (http://plants.ensembl.org/Triticum_aestivum/Tools/Blast) against the TGACv1 wheat genome assembly using the following parameters e-value $\leq$ 1e-10 and identity $\geq$ 85%. The chromosomal location of durum wheat NAC was compared to those from AA and BB sub-genomes of common wheat retrieved from iTak database. Moreover, homeologous genes identification was carried based on phylogenetic analysis, chromosomal location and sequence homology.

2.4. Public RNA-seq based expression analysis of TtNAC genes

For expression analysis of TtNAC genes in vegetative tissues and under stress conditions, publicly available RNA-seq data for each TtNAC were retrieved from the WheatExp database (Pearce et al., 2015). Traes_id for each gene was obtained by BLASTN (e-value cutoff of 1e-10) considering only alignments with minimum 75% sequence coverage and 90% sequence similarity against wheat transcriptome. FPKM (Fragments Per Kilobase Of Exon Per Million Fragments Mapped) expression values during developmental time course in five tissues (Choulet et al., 2014), under drought and heat stresses of a tolerant wheat cultivar (Liu et al., 2015) were retrieved.

2.5. Plant material and stress treatments

Two Tunisian cultivars of durum wheat *(Triticum turgidum* L. ssp. *durum),* Mahmoudi (salt-sensitive) and Om Rabiaa (salt-tolerant), supplied by INRAT (Tunis, Tunisia), were used for expression analysis of ten selected NAC genes. Seeds were sterilized in 0.5% NaOCl for 15 min, washed three times with sterile water and then placed on Petri dishes with a single sheet of Whitman #1 filter paper for germination. Four-day-old seedlings were transferred to tubes containing half-strength Hoagland's solution. All seedlings were grown in a glasshouse at 25 ± 5 °C, under photosynthetically active radiation of 280 μmol m^{-2} s^{-1}, a 16 h photoperiod and 60± 10% relative humidity. Stress treatment was performed on 14 days aged plantlets by supplementing the nutrient solution with NaCl (100 mM) and PEG6000 (15%) for 2, 4, 6, 24 and 48 hours. Leaves and roots from each time point were collected, frozen immediately in liquid nitrogen and stored at −80 °C until RNA isolation. Three biological replicates were carried and each replicate sample consisted of pooled tissues from three plants cultivated under the same conditions.

2.6. RNA isolation and quantitative RT-PCR analysis

Total RNA isolation was performed on leaves and roots tissues using TRIzol reagent following manufacturer's instructions. Total

RNA was DNase-treated using 1 U of RNase-free DNase (Thermo Fisher Scientific) for 20 min at 37 °C and then used for reverse transcription using M-MuLV reverse transcriptase (Thermo Fisher Scientific). The efficiency of reverse transcription was assessed using wheat actin gene (AB181991.1) mRNA pairs of primers (Supplementary Table 1). cDNA samples were diluted five-fold and stored at −20 °C until qPCR analysis. The primers for RT-qPCR (Supplementary Table 4) for each TtNAC gene, excepting TtNAC-B35, were designed in the C-terminal region with melting temperatures of 60 °C using the Primer3 v.0.4.0 software. All quantitative RT-PCR reactions were performed in a CFX96 Real-Time PCR Detection System (Bio-Rad) as follows: initial step of 2 min at 50 °C and 10 min at 95 °C, followed by 40 cycles of 15 s at 95 °C, and 1 min at 60 °C. PCR product specificity was confirmed by melting-curve analysis to ensure that PCR reactions were free of primer dimers and non-specific amplicons. Triplicate experimental sets of qPCR reaction samples were made including the reference control genes and negative controls in a final volume of 20 μl containing 1 μl of cDNA template, 2 × SYBR Green PCR Master Mix (Applied Biosystems, CA, USA), and 300 nM of forward and reverse primers. The efficiency of each pair of primers was determined by plotting the Ct values obtained for serial dilutions of a mixture of all cDNAs and the equation Efficiency = $10^{(-1/\mathrm{slope})} - 1$. Two gene encoding ADP-ribosylation factor (Ta2291) and cell division control protein (Ta54227) which were previously reported as the most stable reference genes to normalize gene expression in different tissue and development stages of wheat (Gimenez et al., 2011) were used for data normalization using formula 3 as described (Pfaffl, 2001) and adopting a pool of all samples as calibrator.

3. Results

3.1. Identification of NAC domain genes in T. turgidum

Deduced protein sequences of *T. turgidum* (Krasileva et al., 2013) were analyzed using iTAK software version 1.6 leading to the identification of 194 putative NAC proteins. The resulting protein sequences were subjected to Pfam analysis (http://Pfam.sanger.ac.uk/) to confirm the presence of the corresponding conserved NAM domains (PF02365). Sequences with incomplete NAM domain were further removed. This process yielded to the identification of 168 non-redundant candidates proteins (Supplementary Table 2) including three previously described NAC genes TtNAM-A1, TtNAM-B1 and TtNAM-B2 (Uauy et al., 2006). Among the new identified proteins, 92 proteins were in full length while the remaining sequences were partial in N-terminal or/and C-terminal region but containing complete NAC domain. TtNACs in the A sub-genome were named from TtNAC-A1 to TtNAC-A84 while those in the A sub-genome were named from TtNAC-B1 to TtNAC-B84 according to their chromosomal location.

The number of NAC proteins is positively correlated with the size and the degree of ploidy of plant genomes. Search for NAM domain containing proteins in triticeae species available in the Ensembl Plants database (http://plants.ensembl.org/) showed that compared to other triticale species durum wheat had more NACs than *Triticum urartu* NAC (103 NACs), *Aegilops tauschii* (117 NACs) and barley (48 NACs). In contrary durum wheat possessed less NAC protein that bread wheat (246 NACs from iTAK database). Compared to other monocots species, durum NAC possessed higher number of NACs than rice (144 NACs) and sorghum (113). Besides ploidy and genome size, expanding NAC genes in durum wheat compared to other cereals and monocots could be explained by gene duplication event to fill a definite biological function as the adaptation to climatic conditions.

We have also compared TtNACs against those of bread wheat in the plants Ensembl database (TGACv1 assembly considering homology ≥ 75%) by BLASTP led to the identification of 148 bread wheat homologs mainly distributed in the A and B sub-genomes (Supplementary Table 2). Furthermore, TtNAC genes harbor an overall homology of 97% with those from bread wheat which suggested high reconciliation between the two species NAC proteins. Besides, 139 TtNACs harbor homology higher than 95% with their bread wheat counterparts.

3.2. Phylogenetic classification and structural analysis of TtNAC proteins

The evolution of the TtNAC proteins was evaluated through phylogenetic analysis incorporating well-characterized NAC genes (Supplementary Table 3) from dicots: Arabidopsis and grape (Shen et al., 2009) and monocots: rice (Shen et al., 2009) and barley (Christiansen et al., 2011). TtNAC proteins were distributed into eight phylogenetic groups (Fig. 1; Supplementary Table 2). This classification revealed good agreement with previous classification based on the phylogenetic analysis of NAC transcription factors from eleven plants species (Shen et al., 2009).

Phylogenetic groups NAC-a to NAC-d contained well-characterized NAC genes from Arabidopsis and rice which have been shown to be involved in the response to biotic and abiotic stresses. Among these genes, OsNAC4, OsNAC5 and OsNAC6 are involved in the response to drought, salinity and cold in rice (Puranik et al., 2012). The group NAC-a contained Arabidopsis ATAF1 and ATAF2 genes and rice OsNAC3-6 which are involved in the response to various biotic and abiotic stresses as well as signaling pathways involving phytohormones (Puranik et al., 2012). In addition, this group contained the barley HvNAC6 that affects the accumulation of ABA and improves resistance against mildew (Chen et al., 2013). This group includes 21 TtNACs (Table 1). Through Phylogenetic analysis, two homeologous genes (TtNAC-A51 and TtNAC-B51) were identified as homologs of rice ONAC002 (sNAC1) involved in stomatal closure and improve drought and salt tolerance in rice (Hu et al., 2006). Similarly, other homeologous genes TtNAC-A60 and TtNAC-B60, TtNAC-A5 and TtNAC-B5 were identified as homologs of OsNAC3 and OsNAC6 genes, respectively. These latter are involved in the response to multiple stresses in rice (Fang et al., 2008).

The group NAC-b contained 17 NAC from durum wheat (Table 1). This group also contained 18 membrane-bound NAC transcription factors from Arabidopsis and rice (Supplementary Fig. 1). Membrane-bound NAC transcription factors are known to be involved in many abiotic stresses response in Arabidopsis and rice (Kim et al., 2010). Indeed, the prediction of trans-membrane helices in TtNAC proteins, allowed the identification of nine putative NTL proteins (NAC with Transmembrane Motif 1-like) in durum wheat (TtNAC-B35, TtNAC-B50, TtNAC-A50, TtNAC-B69, TtNAC-A79, TtNAC-A39, TtNAC-B59, TtNAC-A68 and TtNAC-A59). Phylogenetic analysis of durum wheat NTL genes with those from Arabidopsis and rice (Fig. 2B) showed that these genes split into four clades as described (Kim et al., 2010). Furthermore, from these clades homologs of rice NTL6 (TtNAC-B59, TtNAC-A59), NTL5 (TtNAC-B69), NTL4 (TtNAC-B50, TtNAC-A50), NTL3 (TtNAC-B35) and NTL2 (TtNAC-A68) genes were identified.

The Arabidopsis AtNST and AtVND proteins associated with secondary wall formation during fiber development and xylem differentiation are grouped into the NAC-c group, while NAC genes involved in organ initiation and differentiation (CUC) seem to be grouped into the NAC-d group. Groups NAC-c and NAC-d contained 19 and 44 TtNAC genes, respectively. Moreover, 83 genes from monocots versus 25 from dicots plants belonged to NAC-d group. Based on the tree topology this group split into ten subgroups

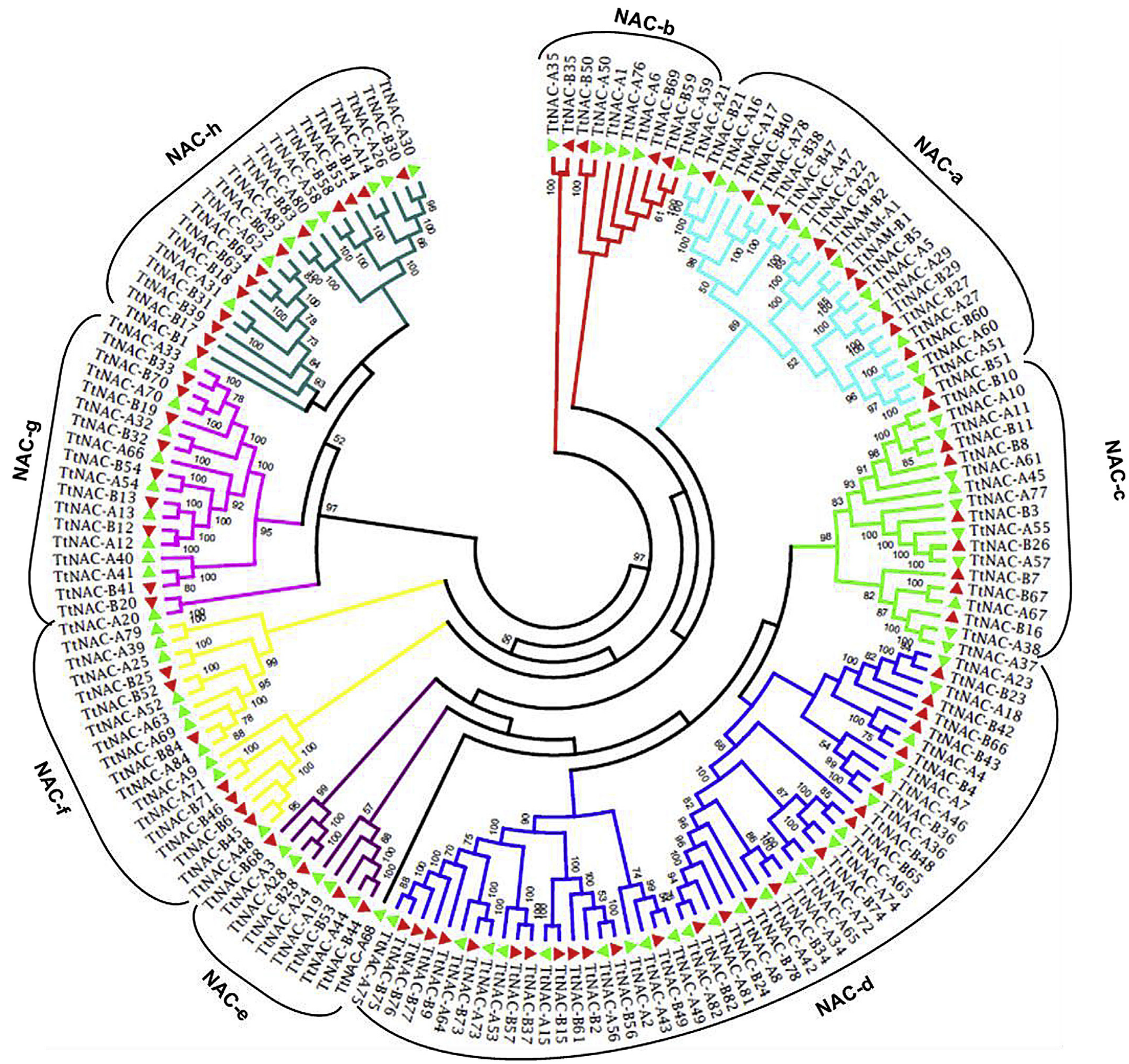

Fig. 1. Phylogenetic relationships of *T. turgidum* NAC proteins. The sequences were aligned by CLUSTALW and MEGA5 was used to construct the unrooted phylogenetic tree deduced by neighbor-joining method. The proteins were classified into eight distinct groups (NAC-a to NAC-h). Each sub-family was assigned a different color according to Shen et al. (2009).

Table 1
The number of NAC genes (in total and specific subgroups) in the 5 plant genomes used for phylogenetic analysis.

Species	NAC-a	NAC-b	NAC-c	NAC-d	NAC-e	NAC-f	NAC-g	NAC-h	Total
T. turgidum	24	17	19	44	15	9	19	21	168
H.vulgare	9	7	3	16	2	4	1	2	44
O. sativa	14	11	10	23	14	10	20	31	133
A.thaliana	12	36	13	16	9	5	14	6	111
V. vinifera	21	13	8	9	15	6	6	0	78
Total	80	84	53	108	55	34	60	60	534

(Supplementary Fig. 1). Only NAC-d-4, NAC-d-5, NAC-d-9 and NAC-d-10 contained NAC genes from Arabidopsis. Indeed, subgroup NAC-d-4 contained only genes from Arabidopsis and grape suggesting that this group was restricted to dicots. The remaining subgroups seemed to be restricted for monocots (Supplementary Fig. 1).

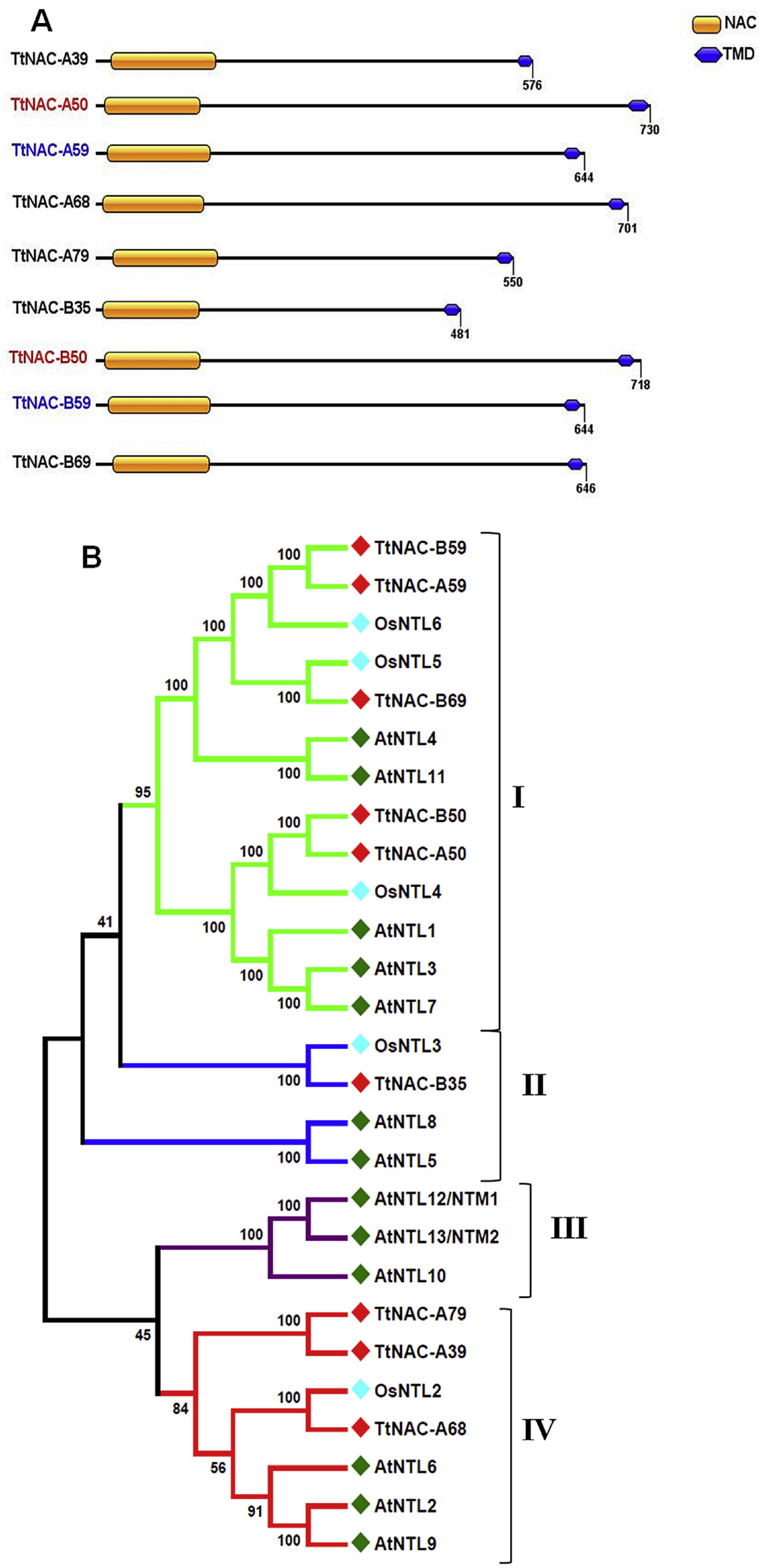

Fig. 2. Putative membrane-bound durum wheat NAC proteins. **A**: Protein structure of membrane-bound NAC TFs. **B**: Phylogenetic relationship of membrane-bound NAC proteins of durum wheat with that of Arabidopsis and rice. Multiple sequence alignment of full-length NAC proteins was done using ClustalW, and the phylogenetic tree was constructed using MEGA5 by the Neighbor-joining method with 1000 bootstrap replicates. The tree was divided into four phylogenetic clades designated by different colors. Homeologous genes were highlighted with the same color.

3.3. Determination of conserved protein motifs

Motif analysis assists in the identification of conserved and novel motifs in proteins. To further examine the structural features of durum wheat NAC genes, conserved motifs were analyzed. Twenty conserved motifs were predicted using the MEME program (Table 2, Supplementary Fig. 2). All TtNACs contained at least one of the four main motifs (motif 1, 2, 3, 4 and 5) that were annotated as NAC sub-domains. Indeed, most of the conserved motifs located in the N-terminal of NAC proteins were highly conserved for DNA-binding. Among the identified motifs, motif 9, 12, 14, 18 and 19 had the minimum sites number (Table 2). Motif 9, 12 and 18 were present only in 5 NAC genes (TtNAM-B2, TtNAC-B22, TtNAC-A22, TtNAM-B1 and TtNAM-A1). These genes belonged to NAC-a group which contained previously characterized stress responsive genes from Arabidopsis and rice. Inherently, motif 15 and 16 were present in 12 and 10 TtNACs, respectively. These motifs were specific to NAC genes belonging to monocots specific NAC-d-7 subgroup (Supplementary Fig. 1). Moreover, motif 7 and 20 were present only in TtNACs from NAC-g and NAC-c groups which contained NAC genes involved in cell wall biosynthesis in Arabidopsis. Generally, NAC proteins that were clustered in same subgroups shared similar motif composition, indicating functional similarities among members of the same subgroup.

3.4. Determination of chromosomal location and identification homeologous genes

The chromosomal positions of TtNAC genes were determined by BLASTN search against bread wheat genome assembly sequences. *In silico* mapping of TtNACs genes on the bread wheat genomes, showed that 51% and 49% of TtNACs were mapped on the bread wheat sub-genomes A and B, respectively (Fig. 3). The mapping of TtNACs genes indicated an uneven distribution of the durum wheat NAC genes on all the A and B sub-genome chromosomes (Fig. 3). Among the A sub-genome, chromosome 2A and 7A contained the highest number of TtNACs (Fig. 3), while minimum genes were distributed on chromosomes 1A, 4A and 6A. Moreover, in the B sub-genome, chromosomes 2B, 3B and 7B contained the highest number of NAC genes (Fig. 3). Chromosome 1B harbored the minimum NAC genes (three TtNACs).

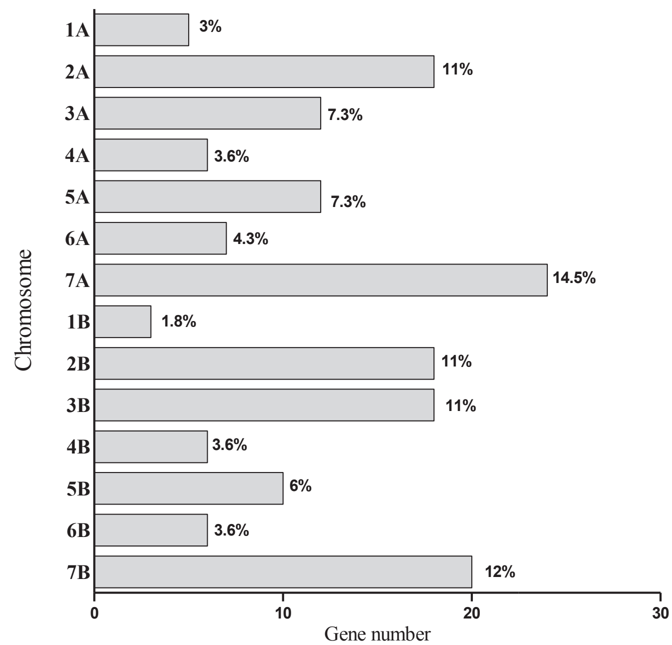

Fig. 3. Distribution of TtNAC genes on bread wheat A and B sub-genomes chromosomes.

The homeologous genes analysis (Supplementary Table 4) led to the identification of 48 homeologous pairs which represented 60% of total NAC genes. Among them, 33 were from stress related phylogenetic groups such NAC-a (18 genes), NAC-b (8 genes), NAC-c (6 genes) and NAC-d (24 genes).

3.5. RNA-seq based expression analysis of TtNAC genes

RNA-seq analysis was employed to investigate the expression pattern of TtNAC genes under normal growth and stress conditions.

Table 2
List of conserved motifs identified in durum wheat NAC proteins.

Motif	Sites	e-value	Amino acid sequence composition	Width (aa)	NAC subdomain
Motif 1	112	6.9e-3285	WYFFSP[RK]DRKYP[TN]G[TI]R[TP]NRA[TA]G[SA]G[YF]WKATGKD[KR][PE][IV]FS	37	C
Motif 2	129	9.9e-1728	G[RK]AP[KR]GE[KR]T[DN]W[VI]MHEYRL	18	E
Motif 3	144	1.1e-1924	PPGFRFHPTDEEL[VI]X[HY]YLR[RPK]K	21	A
Motif 4	132	6.1e-1441	[IAV]IA[ED]VDL[YN][KR]X[ED]PW[DE]LPE[KR]AK[IM]	21	B
Motif 5	148	4.2e-1039	XGRL[VI]GM[KR]KTLVFY[KR]	15	D
Motif 6	142	1.9e-816	SXXK[ED][DE]WV[LV]C[RK][VI][FY]KK	15	–
Motif 7	15	3.7e-421	I[DS][ED]FI[PK]T[IGV][ED][EG][EAG][DA][GR][IF][CG][YC]THP[ER][KND][LM]PG[VMT][KRT] [KMQ]DG[LSH][SA][KLR][HY]FFH[RK][PIK]S[NK][AL]Y[TN][TNV][GE][TQN][RD]K[RD][RI]	50	–
Motif 8	10	1.2e-206	[LF]M[DGE]Y[GD][AM]SS[ST][LC][AV][PH][PA][SPL][SL][VFL]P[TAG][SA]S[SL]Y[QK][LS]H[DA] [VT]G[AT][GR][SL]S[MV]MGS[AV][VA]LP[MI]MN[DNY]HY[FI][GR]N[HN]	50	–
Motif 9	5	5.2e-136	TKQFLAPSSSTPFNWLDAS[TP][VA]GILPQARNFPGFNRSRNVGNMSLSSTADM	50	–
Motif 10	30	9.2e-134	[GK][PG]A[RKS]T[GD]WVMH[EQ]Y[RH][LI][AG]	15	–
Motif 11	31	8.6e-120	RT[RV]XX[SA]E[GT]RWH[KA]TGK[ET][RK]P[VI]X[VD]	21	–
Motif 12	5	7.3e-118	INKAAAGDQQR[SN][MT]ECEDSVEDAVTAYP[LP]YATAGM[TA]GAGAHGSNY[DA]S[LP][LS][HL][HL]	50	–
Motif 13	38	3.2e-107	H[HL]Q[QV]QQQQQHL	11	–
Motif 14	5	5.5e-095	[ND]AD[LP]PMLMD[SY]PSGA[DI]DFAGASSSTSSAA[LV]P[LV]E[LPA]D[MAT]EH[LR][TS][IR]K[MT] EPP[APQV][PQ][QM]Q[QS]	50	–
Motif 15	12	1.2e-088	E[SC][GN]KQ[LG]T[PS][NG][LP][PS][TI][DA][IT][RT]T[VA]T[AT][IT]N	21	–
Motif 16	10	6.6e-078	S[TI]GLKK[VM]V[MTV][PS]SY[AD][MI]PM[PSY][MT][GPS][MA][GE]	21	–
Motif 17	12	1.7e-073	Q[IM]GAD[QE]G[LF]M[IV]G[VA]EPG[SC][GS][PS][SA]S[IM]VS[QR]ED[VT][LV]A	29	–
Motif 18	5	1.4e-065	N[AT][MV][PNS][PA][FM][MS][NT][HPY]LP[VM]QDGTYHQ[QH]HVILG[AT]PL[AV]PEA	32	–
Motif 19	4	9.5e-061	STS[SW]L[PS][QEH]A[AG]NAPF[HSV]LQ[LV]PA[VA]NPPC[GS][LFI][QE]LP[AP]ANHGM [SP]NM[SC][ST]LQLPAAS[QHK]G[VG]	50	–
Motif 20	16	2.8e-059	[DN][AS][AS][GA]T[DE]W[RD]FLD[KRS][LF][LV][AS][ST][QS][LQ]L[HN][EG]	21	–

Corresponding Traes ID for each TtNAC was obtained by BLASTN (identity ≥ 95%; e-value ≤ e−10) in the WheatExp database. Blast results allowed identifying Traes ID for 122 TtNACs with used blast parameters (Supplementary Table 2). These Traes Ids were used in WheatExp database for expression data retrieval. Expression analysis in five developmental tissues showed tissue-specific expression pattern for several NAC genes (Fig. 4A). These genes could be divided into three major expression groups (group I to group III) which contained genes preferentially expressed in grains, leaves and/or roots and expressed in multiple vegetative tissues, respectively. Group I contained 22 genes that were specifically expressed in wheat grain. From which, 7, 2 and one genes belonged to the monocots specific NAC-d-7, NAC-d-2 and NAC-d-1 group, respectively. Furthermore, these genes shared two specific motifs (motif 8, and 17) in their C-terminal regions (Supplementary Fig. 2). Expression group II contained genes that were highly expressed in leaves and/or roots. Contrarily, those in group III were expressed in multiple tissues except in grain (Fig. 4A).

To gain more insight into the expression patterns of TtNACs under abiotic stress, we investigated their expression under drought and heat stresses. Results (Fig. 4B) showed distinct expression pattern of durum wheat NAC transcription factors under drought and heat stresses. Indeed, according to their expression pattern, these genes could be divided into three major expression groups containing 8, 36 and 16 genes, respectively. Moreover, putative stress-related genes, mainly belonged to NAC-a to NAC-d groups, were distributed in these groups. In group II, which contained genes induced by drought, 10, 5 and one gene belonged to NAC-a, NAC-b and NAC-c groups, respectively. Concerning genes in group III, among 16 genes, 10 belonged to stress related

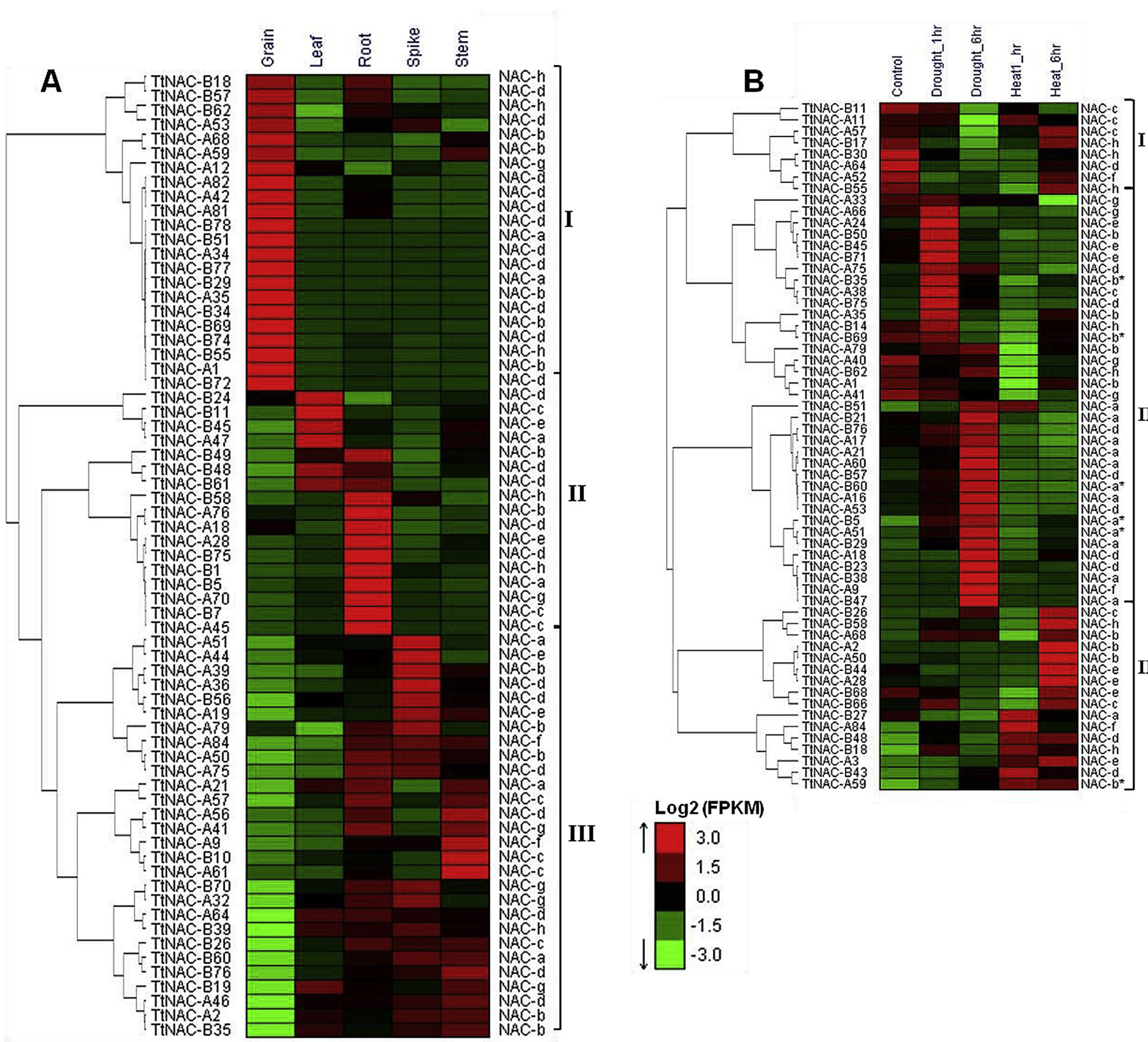

Fig. 4. Heat maps showing RNA-seq based gene expression patterns (FPKM count were Log_2 transformed); **A**: expression patterns during developmental time course in five tissues. **B**: Expression patterns under heat (40 °C) and drought stresses generated from hexaploid bread wheat (var. TAM 107). The colors indicate expression intensity (red, high expression; green, low expression). Expression data were subjected to QT clustering in Expander version 6 software. Gene marked with asterix were selected for RT-qPCR. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

phylogenetic groups and especially from group NAC-d (4 genes).

Homeologous genes identification as described above led to the identification of 48 homeologous pairs. Among them, seven contained genes that were modulated by drought or/and heat stress as revealed by RNA-seq analysis (Supplementary Fig. 3). Expression analysis of pair_4 showed that both homeologs (TtNAC-A11 and TtNAC-B11) were down-regulated by drought and salt while genes in the remaining pairs were up-regulated by both stresses. Three Homeologous pairs (pair_10, 29 and 35) contained genes that reached their maximum expression level after six hours of stresses (Supplementary Fig. 3). Beside, pair_28 and pair_44 contained TtNACs that were induced by both stresses at the first hour. Contrariwise, homeologous genes TtNAC-A50 and TtNAC-B50 showed divergent expression pattern. TtNAC-A50 was up-regulated only under heat stress at six hours while TtNAC-B50 was induced by drought stress after one hour.

3.6. Validation of NAC gene expression by quantitative RT-PCR

Based on phylogenetic analysis and RNA-seq data, a set of ten TtNAC genes were selected to investigate their expression patterns in two durum wheat genotypes under salt (100 mM NaCl) and drought (15% of PEG 6000) stress conditions at different treatment times ranging from 2 to 48 hours. The selected genes, except TtNAC-B33, belonged to stress-related groups. From which three putative membrane-bound NACs (TtNAC-B35, TtNAC-B69 and TtNAC-A59) were the homologs of previously characterized NTL from rice (OsNTL3, OsNTL5 and OsNTL6, respectively). Similarly, five genes (TtNAC-B60, TtNAC-A7, TtNAC-A51, TtNAC-B5 and TtNAC-B27) were the closest homologs of rice stress-responsive NACs (OsNAC3, OsNAC2, sNAC1, OsNAC6 and OsNAC4, respectively). TtNAC-B47 was the closest homolog of Arabidopsis AtNAP while TtNAC-B33, which harbored high protein disorder, was the homolog of ONAC003 (Also known as sNAC3). This latter is induced multiple abiotic stressed and also by abscisic acid in rice (Fang et al., 2015). Quantitative RT-PCR (qRT-PCR) was performed on leaves and roots tissues. Expression analysis under salt stress (Fig. 5) showed that TtNAC-B35, TtNAC-B5 and TtNAC-B27 were not modulated by the salt stress in leaves of the two genotypes while the expression of two genes (TtNAC-B47 and TtNAC-B27) was not affected by salt stress in roots. Moreover, the expression level of TtNAC-B60 was up-regulated in leaves of the tolerant genotype from the first two hours of stress and remained highly expressed until 48 hours of salt stress comparing to the control. It expression in roots showed that TtNAC-B60 was not affected in the sensitive genotype roots while it was induced in roots of both genotypes under salt stress. Indeed, in roots, TtNAC-B60 was induced from two hours in the sensitive genotype while it was induced after four hours under salt stress in the tolerant ones (Fig. 5B). This gene reached the maximum expression after 6 and 24 hours in the sensitive and tolerant genotype, respectively. Moreover, TtNAC-A7 showed similar expression patterns in both leaves and roots. This gene was up-regulated in leaves and roots of the tolerant genotype under both stresses (Fig. 5). Moreover, it reached the maximum expression level after four and eight hours in leaves and roots, respectively. Contrarily, TtNAC-A7 was down-regulated by salt stress in the sensitive genotype and it expression was not affected by salt stress in leaves (Fig. 5A). Two genes (TtNAC-B35 and TtNAC-B27) were induced by salt stress in the roots of both genotypes. These genes were induced from two hours of stress and they remained highly expressed until 48 hours compared to the control. TtNAC-A51 was up-regulated in Om Rabiaa leaves under salt stress white it seemed to be induced only in Mahmoudi genotype roots under salts stress. Concerning TtNAC-B33 and TtNAC-B47, these two genes were repressed in leaves of both genotypes while only TtNAC-B33 was induced by salt stress in Om Rabiaa roots. Moreover, two genes (TtNAC-B69 and TtNAC-A59) showed contrasting expression patterns under salt stress in leaves (Fig. 5A). Indeed, TtNAC-B69 was down-regulated by salt stress in leaves and roots of both genotypes, while TtNAC-A59 was up-regulated by salt stress in Om Rabiaa leaves and roots.

Expression analysis under drought stress (Fig. 6) showed that contrarily to salt stress, TtNAC-B60 expression was not affected by drought in leaves while it was induced under salt stress in the tolerant genotype. Beside, the expression of four NACs (TtNAC-B33, TtNAC-B5, TtNAC-B27 and TtNAC-A59) was not modulated by drought stress in both genotypes roots. Three genes (TtNAC-A7, TtNAC-B47 and TtNAC-B69) showed similar expression patterns in leaves and roots under drought stress. Furthermore, TtNAC-A7 and TtNAC-B47 were induced by drought in both genotypes. They reached their highest expression level after four hours of stress (Fig. 6). Contrarily to TtNAC-A7 and TtNAC-B47, TtNAC-B69 was down-regulated by drought stress in both genotypes. TtNAC-B35 was up-regulated by drought stress in leaves while it was repressed in roots of both genotypes. Indeed, it reached the maximum expression level after four hours of stress. Under salt stress, TtNAC-B35 was induced in roots and not affected in leaves (Fig. 5). The expression of the remaining genes (TtNAC-B5, TtNAC-B27 and TtNAC-A59) was modulated under drought in at least one genotype. TtNAC-B5 was up-regulated in the sensitive genotype leaves while TtNAC-B27 and TtNAC-A59 were induced in the tolerant one (Fig. 6A).

Considered together, expression data from salt and drought stresses showed that the expression of several genes was modulated by both stresses. Their expression was genotype and/or tissue specific. TtNAC-A7 was induced by salt stress only in the tolerant genotype while it was up-regulated by drought in both genotypes. Moreover, TtNAC-B47 and TtNAC-B5 were induced only under drought stress (Fig. 6). Interestingly, TtNAC-A59 seemed to be preferentially induced in the tolerant genotype "Om Rabiaa" under salt and drought. All these genes belonged to stress-related phylogenetic groups. Among them TtNAC-B60, TtNAC-A51, TtNAC-B47 and TtNAC-B27 belonged to the NAC-a group while TtNAC-A7 and TtNAC-B35 belonged to the NAC-b and NAC-d groups, respectively.

4. Discussion

Drought and salinity are two of the main environmental stressors and decrease significantly the crop productivity. Durum wheat like other cereal such as barley and bread wheat are the most cultivated crops in the south and east Mediterranean basin where drought and salinity are the main constraints limiting grain yield. With wheat genome and transcriptome sequenced and the sequencing of several other plant genomes is in progress, the systematic analysis of transcriptional regulation is advancing. These data provided rich resources for comparative genomic analyses. Furthermore, with the rapid development in bioinformatics analyses, the information stored in various plant genomes could be explored to elucidate the mechanisms regulating plant growth, development and multiple stress responses.

As far as we know, compared with Arabidopsis, rice, and other species, a few NAC transcription factors were identified and characterized from wheat species. In bread wheat (*T. aestivum*) a set of NAC transcription factor were characterized under abiotic stresses (Baloglu et al., 2014). In durum wheat (*T. turgidum*) NAC transcription factors plays important role in the regulation of senescence and improves grain protein and some mineral content in wheat (Uauy et al., 2006). Moreover, TtNAMB-2 participates in the responses to various biotic and abiotic stresses (Baloglu et al., 2012). However, the NAC genes family in durum wheat remains

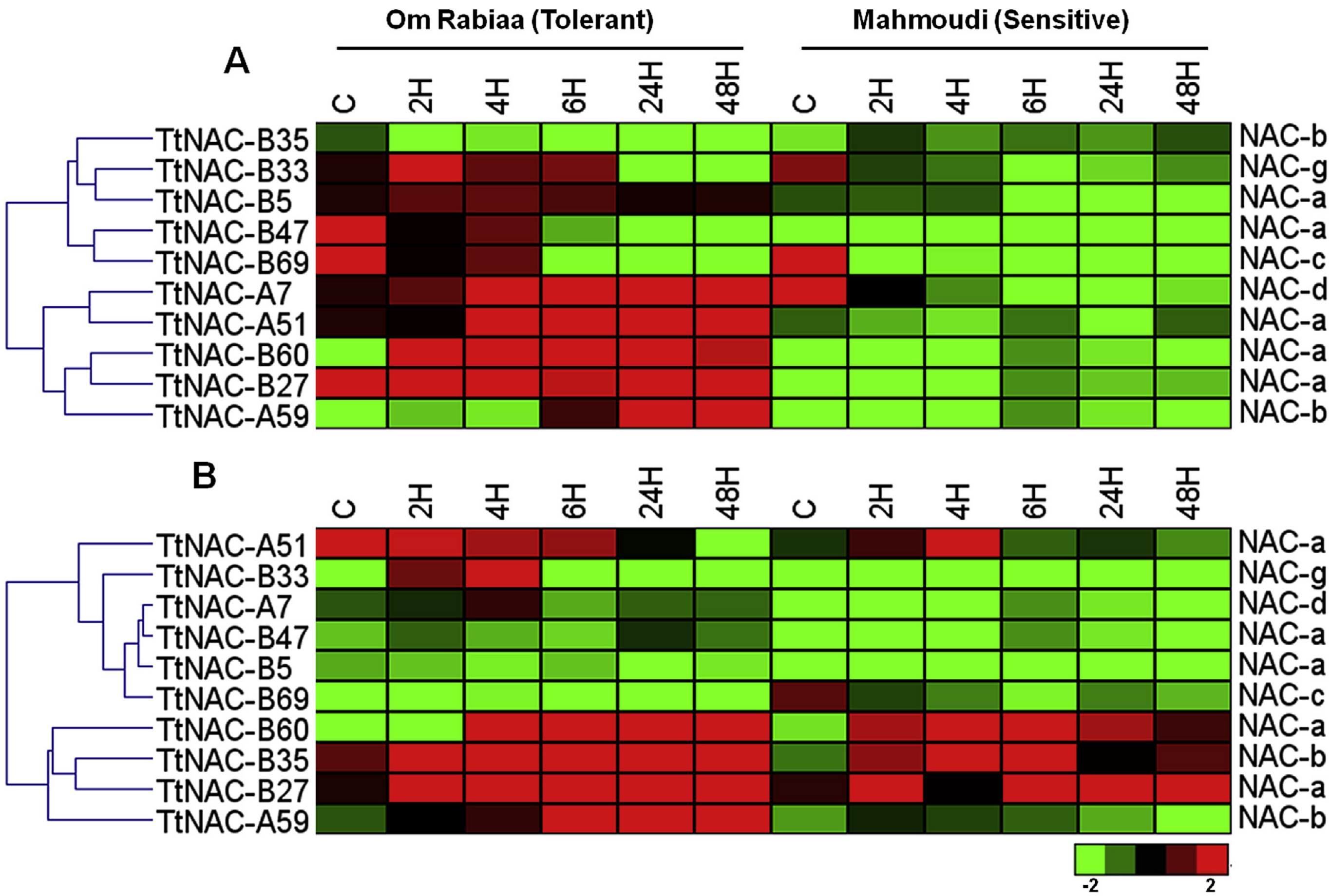

Fig. 5. Expression analysis of selected NAC genes using qRT-PCR in leaves and roots under salt stress (100 mM NaCl) during 48 hours.

uncharacterized. Therefore, it is essential to identify and annotate these genes in durum wheat. In the present study, we identified 168 NAC transcription factors from the whole durum wheat transcriptome including three previously identified members. We characterized the transcription factors expression patterns by *in silico* processing using public RNA-seq data during wheat developmental stages and under drought and heat stresses.

Moreover, we carried a comparison of species homologs of TtNACs with those from monocots and dicots species. Phylogenetic analysis (Table 1; Supplementary Fig. 1) showed that NAC transcription factors from durum wheat were closer to other monocot species used in this analysis. Therefore, we assume that transcription factors within the same taxonomic group exhibit recent common evolutionary origins, and specific conserved molecular functions. We found an expansion of TtNAC proteins in subfamilies NAC-a, NAC-d, and NAC-h. This expansion may be due to tandem duplication events and polyploïdy in wheat. The determination of the chromosomal location of TtNAC against common wheat whole genome assembly revealed an evenly distribution of these TtNAC on the A (51%) and B (49%) sub-genomes. Moreover, we have identified 48 homeologous pairs which represent 60% of all TtNACs.

In these subfamilies homologs of well characterized stress-related NAC transcription factors from rice were identified (Supplementary Table 1). Furthermore, in the NAC-d group, monocots specific phylogenetic clades were identified (Supplementary Fig. 1). The NAC-b group, also contained stress related NACs (Shen et al., 2009), withdrawn 17 TtNAC genes from which 9 genes putatively contained Trans-membrane helices as revealed by *in silico* analysis (Fig. 2). All TtNAC proteins with predicted TMHs, excepting TtNAC-A79 and TtNAC-A39, are homologous to rice TMH-containing genes (Fig. 2). Moreover, two rice genes (OsNTL4 and OsNTL6) processed two durum wheat homologs.

The distribution of amino acid motifs in TtNAC proteins revealed specific conserved motifs through NAC groups (Table 2). For example, Motif 9, 12 and 18 were present only in 5 TtNAC genes belonging to NAC-a group. In between, previously described TtNAM-B1 and TtNAM-A1 that are involved in senescence and grain zinc, and iron contents (Uauy et al., 2006). Moreover, two motifs (motif 15 and 16) were restricted TtNACs belonging to monocots phylogenetic clades.

We have also used available RNA-seq data as useful tool for preliminary analysis of gene expression and identified tissue-specific, stress-responsive TtNAC genes. *In silico* expression during wheat development revealed grain specific expression pattern of monocots specific TtNACs. Moreover, Among NAC genes which were preferentially expressed in grain, 2, 3 and 11 genes belonged to NAC-a, NAC-b and NAC-d groups. In this latter, two homeologs (TtNAC-A34 and TtNAC-B34) were closely related of barley HNAC019 which was suggested as regulator of seed development (Christiansen et al., 2011). Moreover, TtNAC-B74, TtNAC-B72, TtNAC-A53 and TtNAC-B57 belonged to a small wheat-specific group (Supplementary Fig. 1). The *in silico* gene expression presented here (Fig. 4A) suggested that these four genes, together with TtNAC-A34 and TtNAC-B34, could regulate aspects of seed development. This results is consistent with phylogenetic and motif

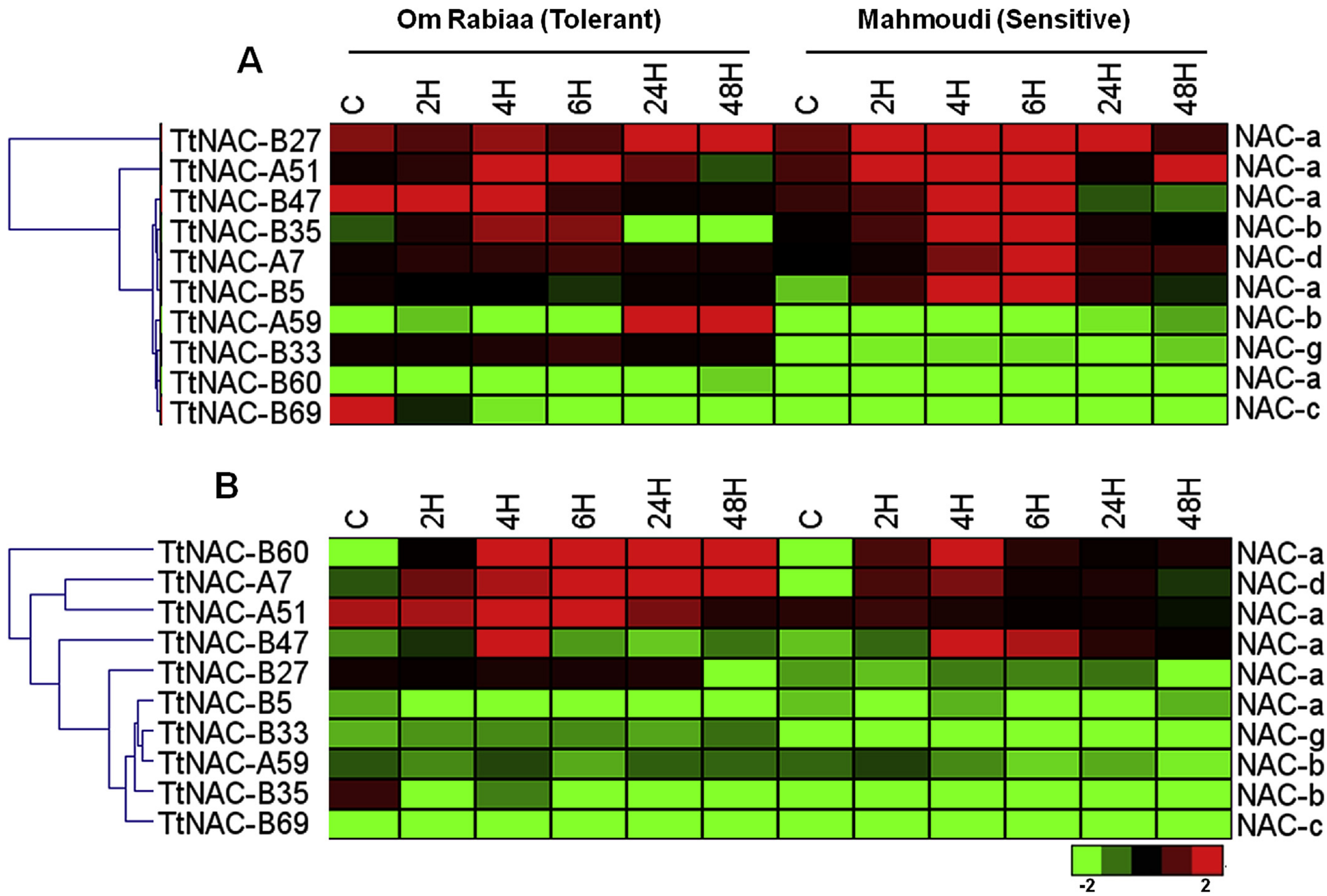

Fig. 6. Expression analysis of selected NAC genes using qRT-PCR in leaves and roots under drought stress (15% PEG6000) during 48 hours.

analyses, genes from NAC-d-7 subgroups shared conserved motifs (motifs 15 and 16) in the C-terminal sequences (Supplementary Fig. 2).

Expression analysis showed that TtNAC-B35 was induced by drought stress in both genotype roots and leaves. TtNAC-A59 was up-regulated by both stresses in the tolerant genotype roots. The expression pattern of these two NACs was consistent with those of RNA-seq data under drought stress (Fig. 4B). Phylogenetic analysis showed that TtNAC-B35 and TtNAC-A59 were the closet homologs of rice OsNTL3 and OsNTL6, respectively (Supplementary Fig. 1). These latter are involved in the response to various abiotic stresses in rice including drought, salinity, cold and heat (Kim et al., 2010). Considering that RNA-seq data were retrieved from leaves of a tolerant wheat cultivar, the expression pattern of TtNAC-A59 corroborates those from *in silico* analysis. Moreover, TtNAC-A59 was also induced by drought stress in leaves after 6 hours (Fig. 4B) while it was induced after 24 hours as revealed by RT-qPCR (Fig. 6). TtNAC-B47 was found to be up-regulated by drought in the two genotypes by RT-qPCR and also after 6 hours of drought stress by RNA-seq. this TtNAC was a close homolog of barley HNAC027 and Arabidopsis AtNAP. This latter is modulated by several stresses such dehydration, salinity and osmotic stress and also by hormones such ABA and ethylene (Guo and Gan, 2006). Its barley homologs (HNAC027) also belonged to group NAC-a, as does the Arabidopsis AtNAP, and highly responded to ABA and methyl jasmonate, suggesting that it could play a role in stress tolerance regulation (Christiansen et al., 2011). All these findings suggested that TtNAC-B47 could be involved in response to stress in wheat mediating hormone signaling pathway. Similarly, TtNAC-B60 was induced by both stresses in the tolerant cultivar roots (Figs. 5B and 6B). Moreover, TtNAC-B60 was the homolog of rice OsNAC3 involved in response to biotic and abiotic stresses (Fang et al., 2008). As for TtNAC-B35, TtNAC-B60 harbor contrasting expression pattern under drought stress in leaves compared to RNA-seq analysis. The TtNAC-B33 gene was found to be up-regulated by salt stress in both tolerant genotype tissues and not by drought stress as revealed by RNA-seq analysis. Moreover, the deduced protein sequence of this gene harbor the highest percentage of disorder in both N-terminal and C-terminal regions (67% overall) among all TtNAC genes. Disordered proteins have also been found in plants in many stress-response processes, acting as protein chaperones or protecting other cellular components and structures (Pazos et al., 2013). Moreover, bioinformatic and experimental studies have revealed that the C-terminal regions of NAC TFs are intrinsically disordered (Jensen et al., 2010). Contrarily to RNA-seq data, the expression of TtNAC-A7 was induced by drought stress in the sensitive genotype leaves while it was highly up-regulated in both cultivars roots under drought and salt stresses. However, expression data obtained for TtNAC-A7 corroborates those reported for its rice homolog OsNAC2 which conferring cold and salt tolerance in rice (Hu et al., 2008). Besides, its barley homolog (HNAC014), which also belongs to NAC-d group, is induced by ABA and methyl jasmonate after 5 hours of treatment (Christiansen et al., 2011). Concerning TtNAC-A51, this gene was found to be induced by drought stress by RT-qPCR and RNA-seq analyses. Moreover, it belonged to NAC-a group, as does its homologs (HNAC014 and sNAC1), and was induced by salt stress in leaves of the tolerant genotype (Fig. 5A). Moreover, sNAC1 is inducible by ABA treatment and enhances drought resistance and salt tolerance in transgenic rice (Hu et al., 2006). TtNAC-B5, homolog of rice OsNAC6, harbor similar expression pattern compared to RNA-seq. Its rice homolog response to hormones and abiotic stresses (dehydration, salinity and cold) and also to *Magnaporthe grisea* (Nakashima et al., 2007). Considered together, RNA-seq and RT-qPCR data revealed that six NAC genes (TtNAC-A7, TtNAC-A51, TtNAC-B47, TtNAC-B69, TtNAC-B27 and TtNAC-A59) showed similar expression pattern in both analyses while the remaining ones had conflicting expression within these analysis.

In conclusion, we conducted a comprehensive search for NAC transcription factors throughout the durum wheat transcriptome, and we identified 168 members. In addition, comparative phylogenetic analysis with Arabidopsis, grape, barley and rice NAC TFs. Phylogenetic analysis allowed the identification of eight NAC groups as previously described. This analysis provides further evidence of the existence of monocots specific NAC members. We have also used available RNA-seq based expression data as a useful tool for preliminary analysis of gene expression and identified tissue-specific and/or stress-responsive TtNAC genes. These founding strongly suggest that preliminary expression profiling using available expression data followed by its validation using RT-qPCR provide more reliable expression analysis of members of large gene families in less time with reduced expenditure.

These data will benefit further investigation of NAC mediated signaling transduction pathways in durum wheat. Overall, this work provides a solid foundation for functional investigation of the NAC family in the response to stress for crop improvement.

Competing interests

The authors' declare that they have no competing interests.

Author Contribution Statements

Mohammed Najib Saidi participated in the design of this work, carried out the experiment, and wrote the manuscript. Dhawya Mergby contributed to carrying out the experiment. Faiçal Brini reviewed the manuscript. All authors read and approved the final manuscript.

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Appendix A. Supplementary data

Supplementary data related to this article can be found at http://dx.doi.org/10.1016/j.plaphy.2016.12.028.

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